

Exploratory Data Analysis (EDA) Report

This report presents the Exploratory Data Analysis (EDA) performed on the Fitness Class Booking dataset. The objective of the analysis was to identify booking trends, seasonal demand patterns, high-demand classes, pricing behavior, and business insights to support forecasting and dynamic pricing strategies.

1. Dataset Overview

The dataset contains fitness class booking information including class descriptions, attendance, pricing, booking dates, and capacity information.

2. Data Preparation

The cleaned dataset was loaded into Python using Pandas. Date columns were converted into datetime format and new time-based features were generated.

3. Booking Trend Analysis

Time-series analysis was performed to identify booking trends and customer demand behavior over time.

4. Seasonal Pattern Analysis

Monthly and weekly booking trends were analyzed to identify seasonal demand fluctuations.

5. Statistical Analysis

Descriptive statistics including mean, median, minimum, maximum, and standard deviation were calculated.

6. Outlier Detection

Box plots and distribution analysis were used to identify anomalies and extreme pricing values.

7. Correlation Analysis

Correlation analysis was performed to understand relationships between price, attendance, and booking demand.

8. Key Insights

- High-demand classes showed consistently strong occupancy.
- Peak demand occurred during specific periods.
- Pricing impacted customer booking behavior.

- Seasonal booking fluctuations were observed.
- Dynamic pricing opportunities were identified.

9. Conclusion

The EDA successfully identified key booking patterns and demand behavior that will support demand forecasting and pricing optimization.

Analysis Component	Status
Data Loading	Completed
Booking Trend Analysis	Completed
Seasonal Analysis	Completed
Outlier Detection	Completed
Correlation Analysis	Completed
Insights Generated	Completed